

Algebraic Topology: a beginner's course

This lecture:

**AlgTop21: The two-holed torus
and 3-crosscaps surface**

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Lecture 21: The two-holed torus + 3-crosscaps
; surface



Lecture 21: The two-holed torus + 3-crosscaps ; surface

There are a lot more symmetries
in hyperbolic geometry!

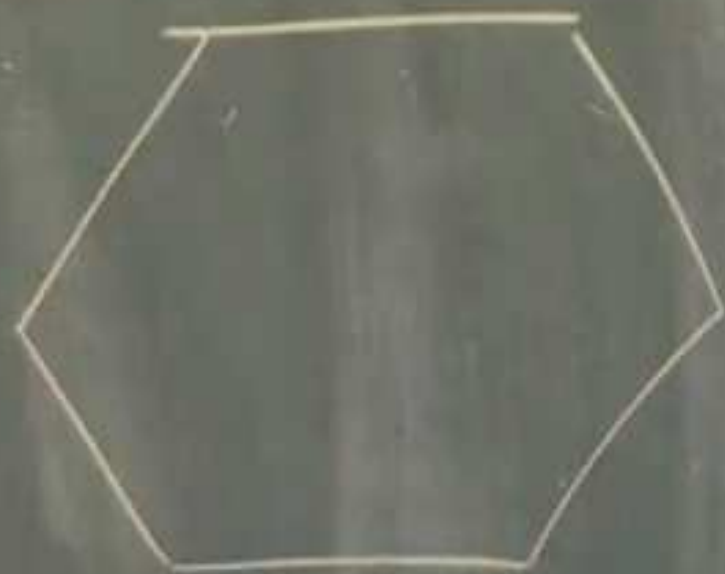
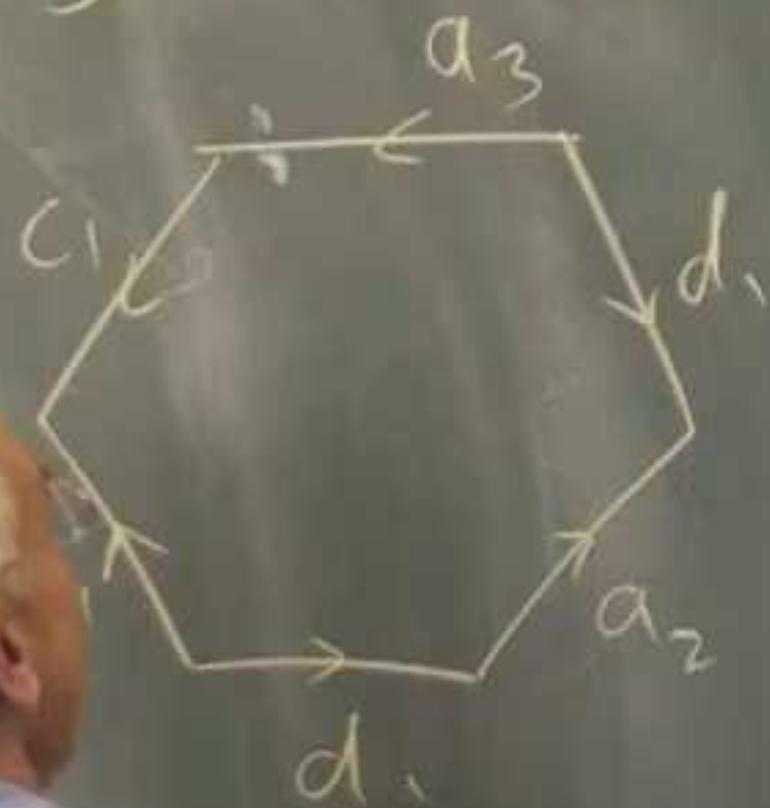
⇒ many more interesting tessellations

Spherical: Platonic solids + ...

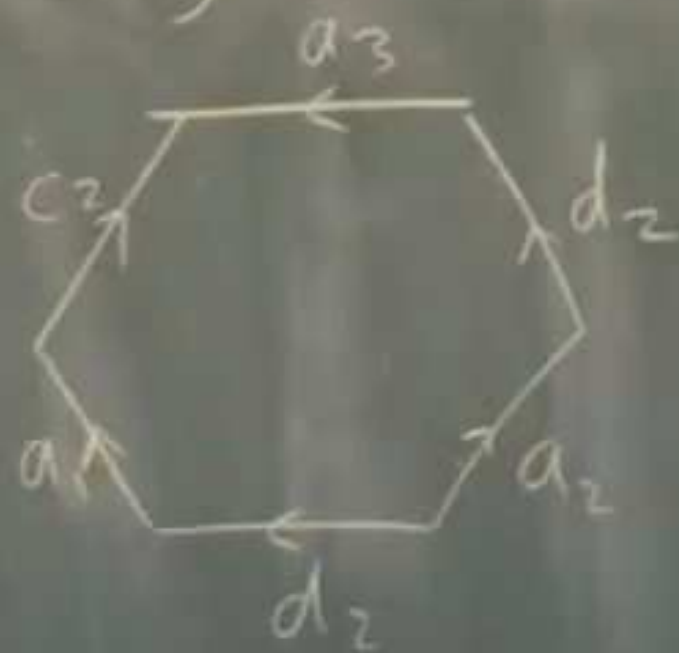
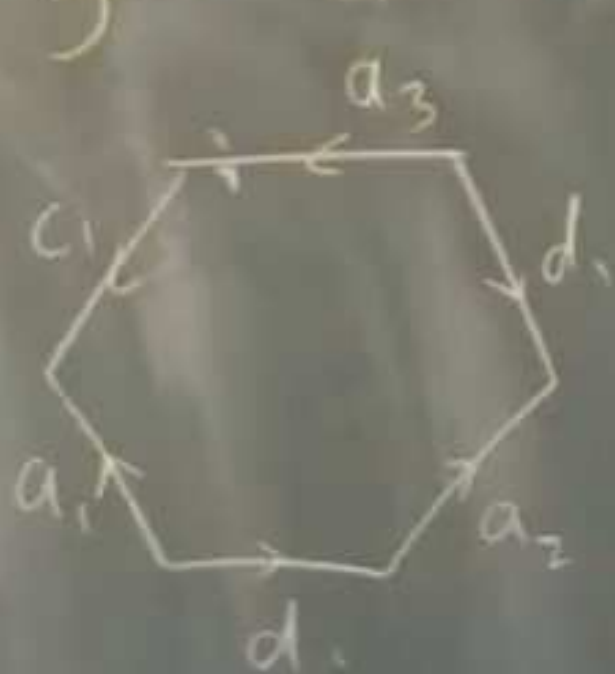
Eucl: 

Hyperbolic: lots

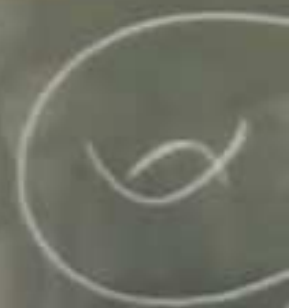
get 4 hexagons:



get 4 hexagons.



Con

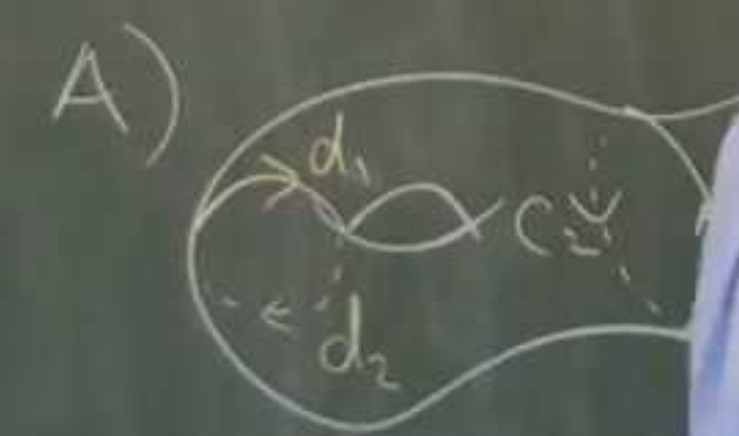


ed sums:

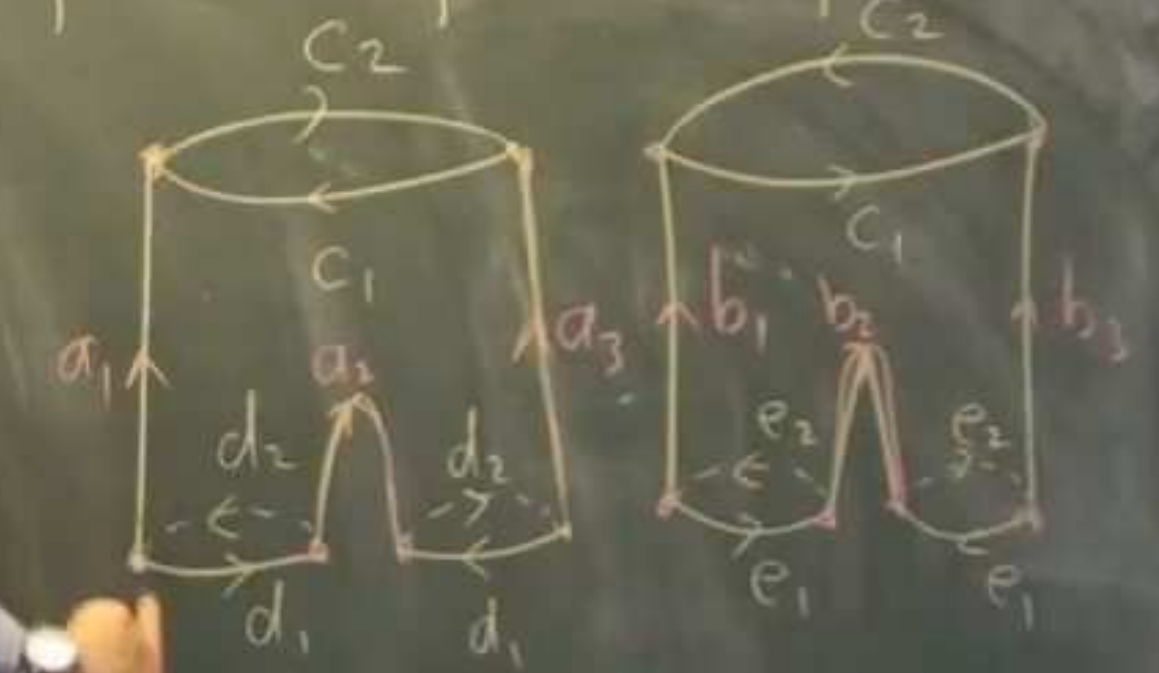
eg. $K = P \# P$

- pair of (pairs of points)

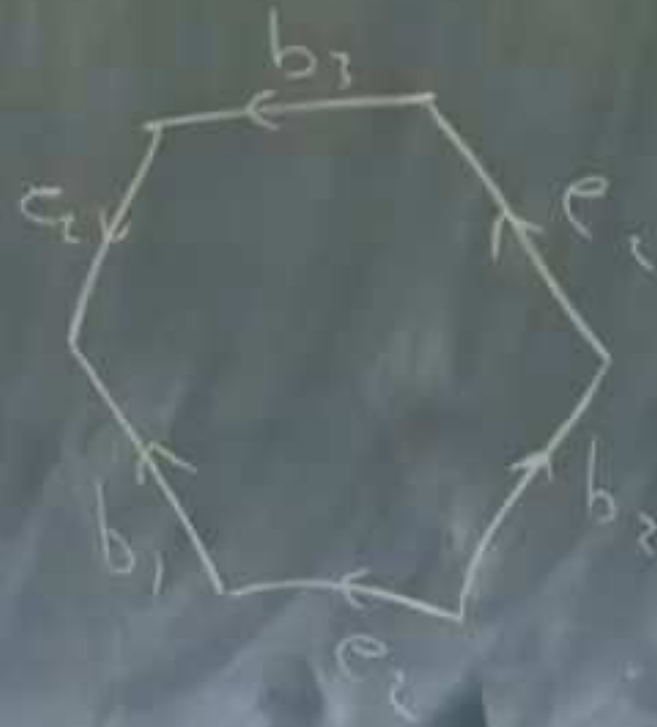
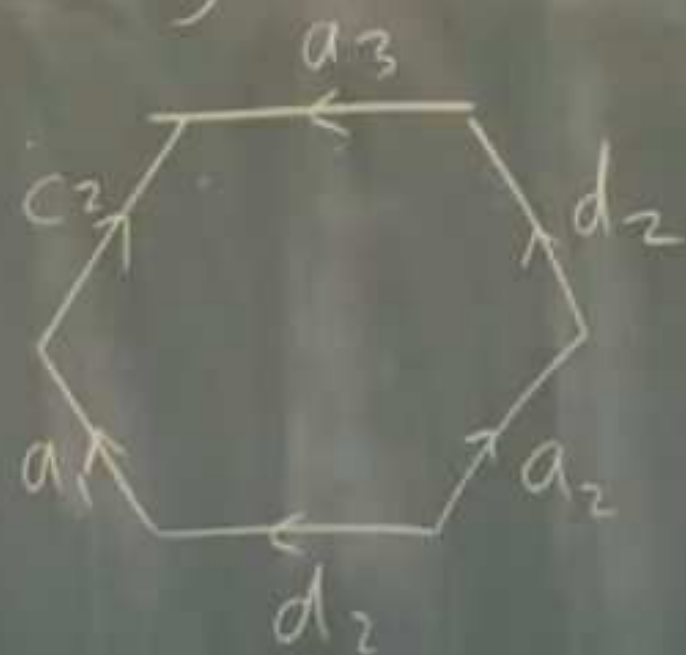
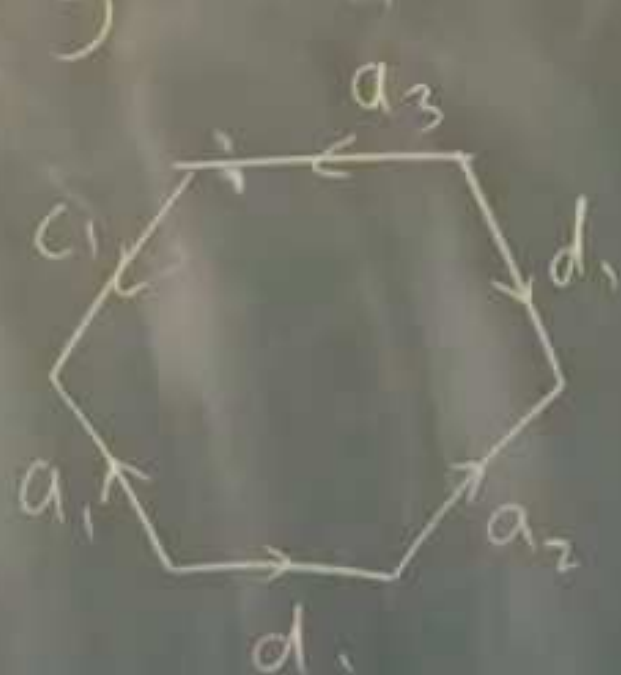
- Thm. 1) S^2
 2) $P \# P \# \dots \# P$
 3) $T \# T$



Two-holed



get 4 hexagons:

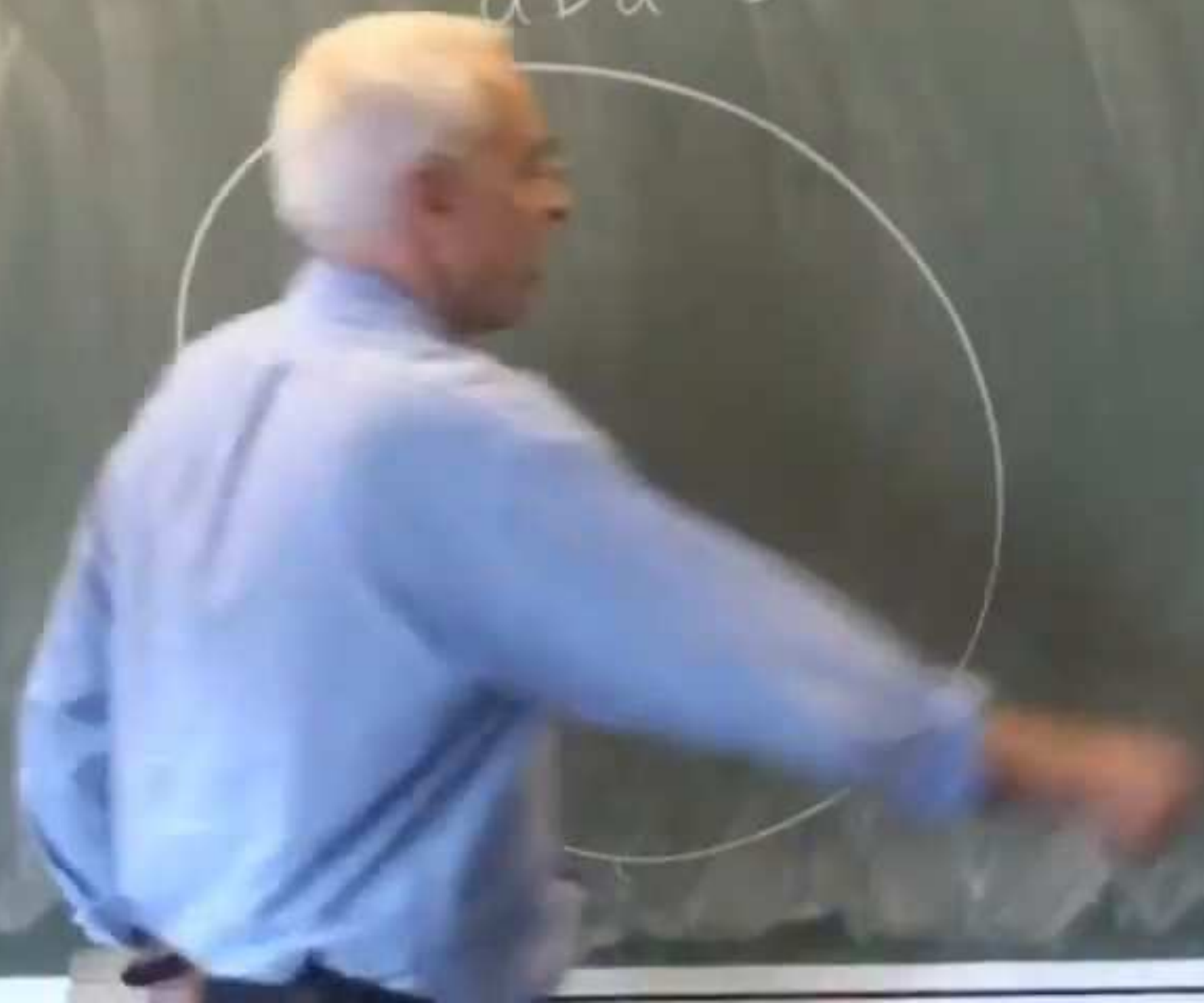


To put
together
these so
four hexagons
meet at a
point.

want regular
hexagons with
angles of $\frac{1}{4}$ at
each vertex.

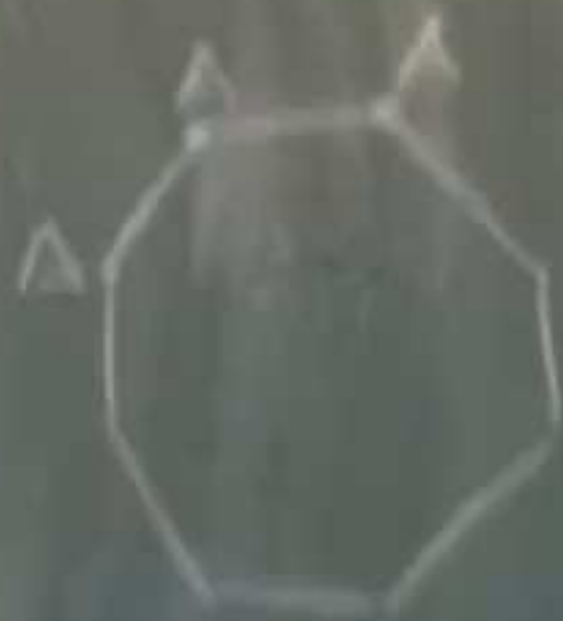
Standard form for $T \# T$.

$aba^{-1}b^{-1}cdc^{-1}d^{-1}$



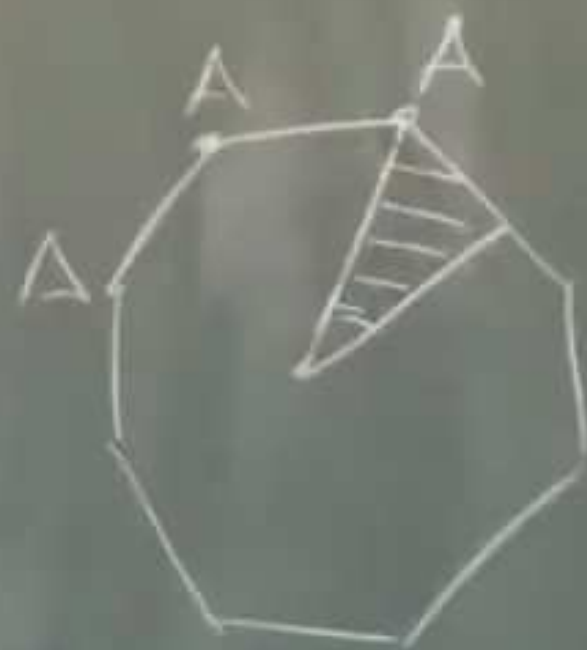
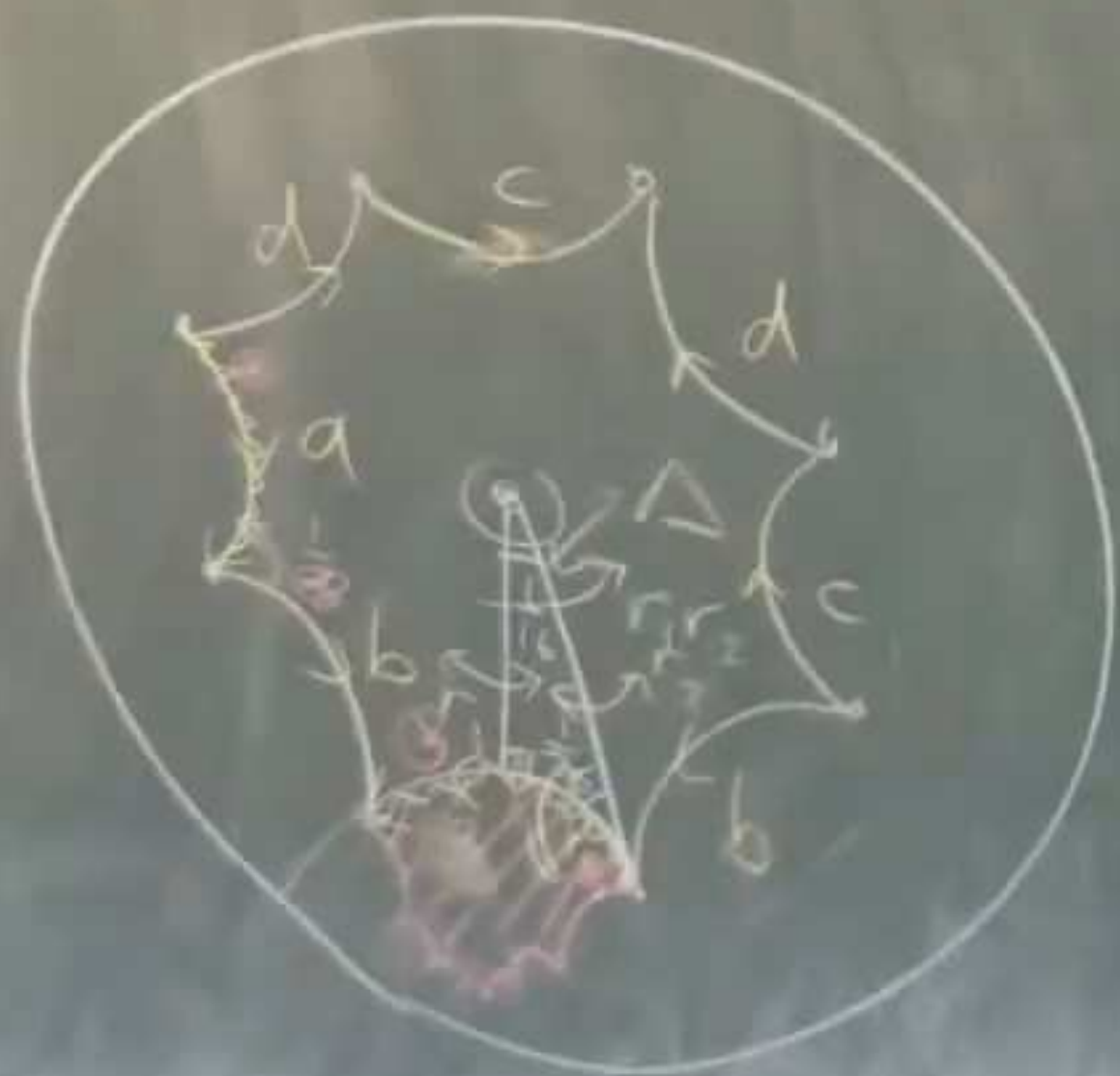
Standard form for $T \# T$.

$a b a^{-1} b^{-1} c d c^{-1} d^{-1}$



Standard form for $T \# 1$.

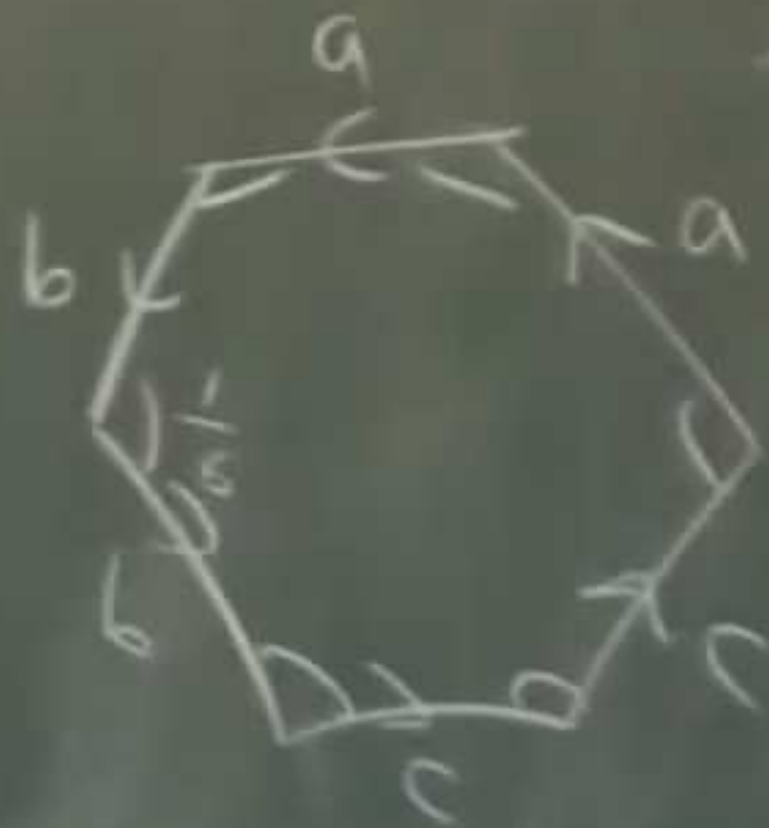
$$aba^{-1}b^{-1}cdc^{-1}d^{-1}$$



$T_a: r_3 r_1 (r_1 r_2)^2$
also need T_b, T_c, T_d

$P \# P \# P$

standard form. $aabbcc$



T.



3 crosscap surface

$P \# P \# P$

standard form. $aabbcc$

Esch



